



# LORENZO LAZZARINO

He/Him | lazzarino@maths.ox.ac.uk | lorenzolazzarino.eu

PHD STUDENT

University of Oxford – STFC



I am a PhD student in Numerical Mathematics at the University of Oxford, advised by Yuji Nakatsukasa and Hussam Al Daas (STFC–CMG). My research concerns (randomized) numerical multilinear algebra, with particular interests in randomized low-rank approximations, a-posteriori error estimates, matrix perturbation theory, and the design of synchrotron experiments.

## Education

|           |                                                                                                                                                                                                    |                                            |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 2023–2027 | <b>DPhil in Mathematics</b><br>Advisors: Yuji Nakatsukasa (Oxford) and Hussam Al Daas (STFC–CMG).                                                                                                  | University of Oxford, Oxford, UK           |
| 2021–2023 | <b>MSc in Computational Mathematics</b><br>Graduated with honours. Thesis: <i>Numerical approximation of the time-ordered exponential for spin dynamic simulation</i> ; supervisor: Stefano Pozza. | Charles University, Prague, Czechia        |
| 2021–2023 | <b>4EU+ Programme</b><br>Collaboration with Prof. Christian Bonhomme.                                                                                                                              | Sorbonne University, Paris, France         |
| 2017–2021 | <b>BSc in Mathematics</b>                                                                                                                                                                          | Università Milano–Bicocca, Milan, Italy    |
| 2017–2021 | <b>Erasmus+ Programme</b>                                                                                                                                                                          | TU Kaiserslautern, Kaiserslautern, Germany |

## Publications

- [1] **Reducing acquisition time and radiation damage: data-driven subsampling for spectro-microscopy**  
M. Meier, L. Lazzarino, B. Shustin, H. Al Daas, P. Quinn. *Journal of Synchrotron Radiation*, 2026. [\[paper\]](#)
- [2] **Efficient error estimators for Generalized Nyström**  
L. Lazzarino, K. J. Pearce, N. Pritchard. *arXiv*, 2026. [\[preprint\]](#)
- [3] **Preconditioned normal equations for solving discretised partial differential equations**  
L. Lazzarino, Y. Nakatsukasa, U. Zerbinati. *arXiv*, 2025. [\[preprint\]](#)
- [4] **Matrix perturbation analysis of methods for extracting singular values from approximate singular subspaces**  
L. Lazzarino, H. Al Daas, Y. Nakatsukasa. *SIMAX*, 2024. [\[paper\]](#)

## Teaching

|           |                                                                                                       |                      |
|-----------|-------------------------------------------------------------------------------------------------------|----------------------|
| 2024–2026 | <b>Tutor, Numerical Linear Algebra</b><br>Lecturer: Y. Nakatsukasa.                                   | University of Oxford |
| 2026      | <b>Mentor, Directed Reading Programme</b><br>Reading programme based on Pearce and Martinsson (2025). | University of Oxford |
| 2023      | <b>Marker, Numerical Linear Algebra</b><br>Lecturer: J. Tanner.                                       | University of Oxford |
| 2024      | <b>Marker / Teaching Assistant, Continuous Optimization</b><br>Lecturer: C. Cartis.                   | University of Oxford |

## Awards

|                                                                                                                         |                                                          |                                                        |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------|
| 2026                                                                                                                    | <b>Finalist, STEM for BRITAIN</b>                        | House of Commons, UK Parliament                        |
| Recognised for work on data-driven experimental design in spectro-microscopy.                                           |                                                          |                                                        |
| 2023–2027                                                                                                               | <b>Ada Lovelace Centre Scholarship</b>                   | Mathematical Institute, University of Oxford           |
| 2022                                                                                                                    | <b>Scholarship for Outstanding Academic Achievements</b> | Charles University                                     |
| 2022–2023                                                                                                               | <b>Student Faculty Grants</b>                            | Faculty of Mathematics and Physics, Charles University |
| Team production of Functional Analysis lecture notes in Czech, English, and Italian using a shared mathematical script. |                                                          |                                                        |

## Academic Activities

|                               |                                                                |                      |
|-------------------------------|----------------------------------------------------------------|----------------------|
| 2026–2027                     | <b>Co-organizer, Numerical Analysis Group Internal Seminar</b> | University of Oxford |
| Co-organizer with Kars Knook. |                                                                |                      |

## Talks

|                |                                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 04 Jun 2026    | <b>Error estimations for randomized low-rank approximations</b><br><i>Numerical Analysis Group Internal Seminar</i>   University of Oxford                                           |
| 22 May 2026    | <b>A-posteriori error estimates for randomized low-rank approximations</b><br><i>ILAS 2026</i>   Virginia Tech                                                                       |
| 11 May 2026    | <b>Error estimation for randomized low-rank approximations</b><br><i>PhD Showcase</i>   RAL Campus                                                                                   |
| 16 Apr 2026    | <b>Error estimation for randomized low-rank approximations</b><br><i>COMPMATH Seminar</i>   University of Groningen                                                                  |
| 17 Mar 2026    | <b>Reducing acquisition time and radiation damage through data-driven experimental design in spectro-microscopy</b><br><i>STEM for BRITAIN</i>   House of Commons, UK Parliament     |
| 01 Sep 2025    | <b>Extracting accurate singular values from approximate subspaces</b><br><i>ENUMATH</i>   Heidelberg                                                                                 |
| 25 Jun 2025    | <b>Extracting accurate singular values from approximate subspaces</b><br><i>The 30th Biennial Numerical Analysis Conference</i>   University of Strathclyde                          |
| 05 Jun 2025    | <b>Reducing acquisition time and radiation damage: data-driven subsampling for spectromicroscopy</b><br><i>Numerical Analysis Group Internal Seminar</i>   University of Oxford      |
| 20 Feb 2025    | <b>Matrix perturbation analysis of methods for extracting singular values given approximate subspaces</b><br><i>Numerical Analysis Seminar</i>   Charles University, Prague (online) |
| 20–21 Jan 2025 | <b>Extracting accurate singular values from approximate subspaces</b><br><i>Due Giorni di Algebra Lineare Numerica e Applicazioni</i>   Università di Pisa                           |
| 10 Oct 2024    | <b>Extracting accurate singular values from approximate subspaces</b><br><i>PYSANUM</i>   Università di Pisa and Scuola Normale Superiore                                            |
| 29 Jul 2024    | <b>Matrix perturbation analysis of methods for extracting singular values</b><br><i>Gene Golub SIAM Summer School, 3-minute presentations</i>   Quito                                |
| 05 Mar 2024    | <b>Error bound on singular values approximations by generalized Nyström</b><br><i>Numerical Analysis Group Internal Seminar</i>   University of Oxford                               |

